mystery, until 11 years later, when a massive crater 106 miles (170 km) across on Mexico's Yucatan Peninsula was found to date from the time of the extinction.

The scientific consensus is that a massive comet or asteroid struck Earth, producing a blast of radiation and a destructive megatsunami more than 328 ft (100 m) high. The radiation would have killed animals nearby, and the megatsunami would have obliterated coastal regions around the Gulf of Mexico. The main damage, however, would be more gradual. A vast cloud of soot and dust would have spread through the atmosphere, reducing the amount of sunlight that reached Earth's surface for several years. Plants died because they could no longer photosynthesize, and algae in coral reefs also succumbed, disrupting food chains worldwide.

The impact would have also released sulfuric acid into the atmosphere, which produced acid rain, acidifying the oceans and killing off marine life. Around the same time, a huge amount of volcanic activity flooded 193,000 sq miles (500,000 sq km) of southern India with lava, forming the Deccan Plateau and further changing the climate and atmosphere.

The K-Pg event is best known for the extinction of all nonflying dinosaurs. It was also responsible for the death of nearly all four-legged animals (tetrapods) that weighed more than 55 lb (25 kg). An exception were crocodiles, which may have survived because they are ectotherms (cold-blooded animals), able to survive for a long time without food. Dinosaurs were endotherms (warm-blooded animals), with a fast metabolism that demanded regular meals.

Many plant species died because they could no longer photosynthesize, leaving herbivorous dinosaurs with little vegetation to eat, while predatory species starved for lack of prey. In contrast, fungi, which do not depend on photosynthesis, proliferated.